HUARAVITO LTEMS

Technical Appendix

Basal Area Estimation for Hauya elegans Within Permanent Plot LTEMS-P001

Analysis Date: February 2026

Plot: 30 × 30 m (900 m²)

Author: Randall Santamaría Artavia

---

1. Introduction

Basal area (BA) is a fundamental metric in forest ecology, defined as the cross-sectional area of tree stems at breast height (1.3 m) per unit ground area. It serves as a proxy for stand biomass, canopy dominance, and structural maturity. This technical appendix presents preliminary basal area estimates for Hauya elegans (Guayabón/Supara) within the permanent monitoring plot LTEMS-P001, based on field observations conducted in February 2026.

The analysis employs two scenarios—conservative and realistic—to bracket the likely range of basal area values, accounting for observational uncertainty in DBH (diameter at breast height) measurements conducted without formal dendrometry equipment.

2. Methods

2.1 Plot Characteristics

Permanent plot LTEMS-P001 is a 30 × 30 m (900 m²) quadrat established in the western core sector of Huaravito Reserve, characterized by high structural density of Hauya elegans. The plot was selected to capture the zone of maximum H. elegans abundance within the site.

Parameter	Value
Plot Size	900 m² (0.09 ha)
Hauya elegans Count	15 individuals ≥10 cm DBH
Mean DBH (visual estimate)	~40 cm
Large Individuals (>80 cm)	3 individuals (estimated)
Expansion Factor to Hectare	10,000 / 900 = 11.11

Table 1. Plot parameters and H. elegans population summary.

2.2 Basal Area Formula

Individual tree basal area is calculated as:

$$BA = (\pi \times DBH^2) / 4$$

Where DBH is measured in meters and BA is expressed in m^2. Stand-level basal area (m^2/ha) is calculated by summing individual tree basal areas within the plot and scaling to hectare using the expansion factor (11.11 for a 900 m^2 plot).

3. Scenario 1: Conservative Estimate

3.1 Assumptions

- All 15 individuals assumed to have DBH = 40 cm (0.4 m)
- This represents the lower bound of observed size distribution
- Deliberately underestimates contribution of large individuals (>80 cm DBH)

3.2 Calculation

For DBH = 0.4 m:

$$BA = (3.1416 \times 0.4^2) / 4 = 0.126 \text{ m}^2 \text{ per tree}$$

Total for 15 trees:

$$0.126 \times 15 = 1.89 \text{ m}^2 \text{ (in 900 m}^2 \text{ plot)}$$

Scaled to hectare:

$$\mathbf{1.89 \times 11.11 = 21.0 \text{ m}^2\text{/ha}}$$

> **Conservative Estimate: ≈21 m^2/ha of Hauya elegans basal area**

4. Scenario 2: Realistic Estimate

4.1 Assumptions

- 12 individuals at DBH = 40 cm (0.4 m)
- 3 individuals at DBH = 80 cm (0.8 m)
- This distribution reflects field observations of multiple large-diameter emergent trees with prominent buttress roots

4.2 Calculation

Small-Medium Trees (n=12, DBH=0.4 m)

$$BA = 0.126 \text{ m}^2 \text{ per tree}$$
$$0.126 \times 12 = 1.512 \text{ m}^2$$

Large Trees (n=3, DBH=0.8 m)

$$BA = (3.1416 \times 0.8^2) / 4 = 0.503 \text{ m}^2 \text{ per tree}$$
$$0.503 \times 3 = 1.509 \text{ m}^2$$

Total Plot Basal Area

$$1.512 + 1.509 = 3.021 \text{ m}^2 \text{ (in 900 m}^2 \text{ plot)}$$

Scaled to hectare:

$$3.021 \times 11.11 = 33.6 \text{ m}^2/\text{ha}$$

> **Realistic Estimate: ≈34 m²/ha of Hauya elegans basal area**

5. Comparative Context

5.1 Tropical Forest Benchmarks

To contextualize the estimated basal area values, we compare against published ranges for mature tropical forests:

Forest Type	Basal Area (m²/ha)	Source
Mature tropical forest (total)	25–40	Clark & Clark 2000
Old-growth premontane (Costa Rica)	30–45	Lieberman et al. 1996
Secondary forest (<20 years)	10–20	Chazdon et al. 2007
Hauya elegans at Huaravito (conservative)	**21**	This study
Hauya elegans at Huaravito (realistic)	**34**	This study

Table 2. Basal area comparison with tropical forest benchmarks. Highlighted rows indicate Huaravito estimates.

5.2 Interpretation

The estimated basal area range of 21–34 m²/ha for Hauya elegans alone is notable for several reasons:

- Single-species dominance: In most tropical forests, total basal area (25–40 m²/ha) is distributed across dozens of species. At Huaravito, H. elegans alone accounts for basal area values approaching or exceeding total stand basal area in typical mature forests.
- Exceeds secondary forest values: The realistic estimate (34 m²/ha) substantially exceeds typical basal area for secondary forests <20 years old (10–20 m²/ha), supporting the hypothesis of mature or old-growth forest structure.
- Comparable to old-growth stands: The realistic estimate falls within the lower-to-mid range of old-growth premontane forests in Costa Rica (30–45 m²/ha), consistent with structurally mature forest conditions.

These findings suggest that H. elegans functions as a structural dominant within the Huaravito forest core, exerting disproportionate influence on canopy architecture, light interception, and microclimate regulation.

6. Ecological and Microclimatic Implications

6.1 Structural Dominance

The high basal area of H. elegans (21–34 m²/ha representing 60% of stems ≥10 cm DBH in the plot) indicates:

- Physiological dominance: Large-diameter trees with extensive buttress roots suggest long-term competitive success and resource acquisition
- Canopy occupation: High basal area correlates with high leaf area index (LAI), which directly influences light interception and understory microclimate
- Recruitment bottleneck: The dominance of large individuals with limited representation of small size classes (< 20 cm DBH) may indicate episodic recruitment or ongoing gap-phase dynamics

6.2 Microclimate Regulation

The documented exceptional thermal buffering at Huaravito (sustained cooling of -3.7°C relative to regional ambient; spatial gradients up to -0.108°C/m) is likely mediated in part by the structural attributes associated with high H. elegans basal area:

- Canopy closure: High LAI reduces solar radiation reaching the forest floor, minimizing radiative heating
- Evapotranspirative flux: Large-diameter trees with extensive crown volumes contribute substantial latent heat flux through transpiration
- Epiphytic biomass: H. elegans trunks exhibit >70% epiphytic colonization; this biomass increases thermal inertia and moisture retention, contributing to evaporative cooling
- Organic matter accumulation: Dense root systems and litter production from dominant H. elegans contribute to the observed deep organic soil layer (mean 21.8 cm), which insulates against soil heating and fuels evaporative cooling

Thus, the high basal area of H. elegans is not merely a taxonomic curiosity but a functional driver of the site's exceptional microclimatic stability.

7. Limitations and Future Work

7.1 Methodological Limitations

- Visual DBH estimation: No diameter tape measurements were conducted; DBH values are estimated from field observations and photographic records. Formal dendrometry is required to refine basal area estimates.
- Single plot: The 900 m² plot may not be fully representative of site-wide H. elegans distribution. Additional plots in the transition and edge zones would enable spatial basal area mapping.
- No height measurements: Tree heights were not measured; future work should employ clinometers or UAV-based photogrammetry to estimate canopy volume and biomass.
- Incomplete size-class distribution: The census included all stems ≥10 cm DBH but did not systematically quantify regeneration (<10 cm DBH), precluding demographic analysis.

7.2 Recommended Next Steps

- Formal dendrometry: Measure DBH with diameter tape for all stems ≥10 cm DBH; re-calculate basal area with precision estimates
- Height measurement: Use clinometer or laser rangefinder to measure tree heights; calculate aboveground biomass using allometric equations
- Spatial replication: Establish additional 30 × 30 m plots in edge and transition zones to quantify spatial variation in basal area

- Regeneration census: Conduct understory survey (all stems 1–<10 cm DBH) to assess recruitment patterns
- Dendroch ronological sampling: Extract increment cores from 3–5 large H. elegans to estimate age structure and confirm old-growth hypothesis

8. Conclusion

Preliminary basal area estimates indicate that Hauya elegans contributes 21–34 m²/ha within the permanent monitoring plot LTEMS-P001, representing structural prominence comparable to—or exceeding—total stand basal area in many mature tropical forests. This finding supports the hypothesis that the Toyopán Sacred Forest core exhibits old-growth structural characteristics, including:

- High single-species dominance (H. elegans: 60% of stems)
- Presence of large-diameter individuals (>80 cm DBH) with buttress roots
- Basal area values consistent with mature premontane forests (30–45 m²/ha)
- Deep organic soil layers (mean 21.8 cm) indicative of long-term litter accumulation

The structural dominance of H. elegans likely contributes directly to the site's exceptional microclimatic stability, with high basal area driving canopy closure, evapotranspirative flux, and epiphytic biomass accumulation. These attributes collectively enable the documented biogenic cooling effects (sustained -3.7°C; spatial gradients up to -10.2°C).

Formal dendrometric measurements and spatial replication are required to refine these estimates. However, the convergence of basal area, soil characteristics, and microclimatic data strengthens the working hypothesis that Huaravito Reserve harbors a structurally mature forest remnant with functional attributes comparable to primary premontane forests.

References

Chazdon, R. L., et al. (2007). Rates of change in tree communities of secondary Neotropical forests following major disturbances. Philosophical Transactions of the Royal Society B, 362(1478), 273–289.

Clark, D. A., & Clark, D. B. (2000). Landscape-scale variation in forest structure and biomass in a tropical rain forest. Forest Ecology and Management, 137(1-3), 185–198.

Lieberman, D., et al. (1996). Tropical forest structure and composition on a large-scale altitudinal gradient in Costa Rica. Journal of Ecology, 84(2), 137–152.

Randall Santamaría Artavia

Custodian-Director

Huaravito Long-Term Ecological Monitoring Site

Piedades Sur, San Ramón, Costa Rica

February 2026